

AI Failure Museum: Field Checklist

Six failure modes, twelve questions, and what to ask before any model reaches production.

EX.01 • CLASS IMBALANCE

The 99% Accurate Cancer Detector

50,000 healthy scans, 200 malignant. Always predicting "healthy" scores 99.6% accuracy and catches zero cancers.

EX.02 • DATA DRIFT

The Model That Forgot

A new content category has no training weight behind it. Recommendations quietly worsen, week over week.

EX.03 • DATA BIAS

The Biased Hiring AI

A model trained on skewed hiring history repeats the skew, confidently and at scale, until someone checks.

EX.04 • DATA QUALITY

Garbage In, Garbage Out

Missing values, duplicates, an outlier, one mislabeled record: four quiet defects, one much weaker model.

EX.05 • EXPLAINABILITY

The Black Box Loan

A high-performing model with no reason codes fails a compliance audit. SHAP adds reasons, performance intact.

EX.06 • PRIVACY & SECURITY

Share, Anonymize, or Federate

Raw sharing, anonymizing, and federated learning trade usefulness against privacy risk in different directions.

THE CHECKLIST ROOM: TWELVE QUESTIONS BEFORE PRODUCTION

1. Data availability: enough relevant data?
2. Data quality: complete and consistent?
3. Labeling quality: correct, audited labels?
4. Data bias: reflects who is affected?
5. Data imbalance: enough of the rare class?
6. Data drift: monitored vs. training data?
7. Privacy & security: GDPR / HIPAA respected?
8. Integration: sources combined consistently?
9. Versioning: training data reproducible?
10. Scalability: pipeline handles full scale?
11. Explainability: specific reasons available?
12. Real-time: adapts to streaming data?

Every number in the six exhibits above is illustrative, built to make a mechanism visible, not measured from a real dataset. The stakes behind each one are not illustrative: a biased hiring model or an unexplained loan denial affects a real person's income or opportunity.